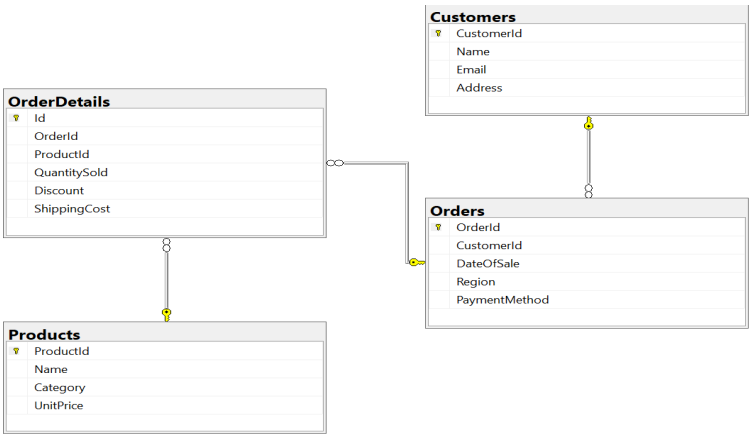
Sales Data API – Design and Implementation Overview

1. Objective

To build a **RESTful API** capable of ingesting sales data from CSV files, storing it in a normalized database, and providing analytical endpoints for reporting top-selling products overall, by category, and by region.

2. Database Design

- **Entities:**
  - **Customer:** Stores customer information (ID, Name, Email, Address).
  - **Product:** Stores product details (ID, Name, Category, Unit Price).
  - **Order:** Represents a sales order (ID, Customer ID, Date of Sale, Region, Payment Method).
  - **OrderDetail:** Contains order-specific product data (Order ID, Product ID, Quantity Sold, Discount, Shipping Cost).
- **Relationships:** Modeled via foreign keys but IDs can be managed as strings to simplify CSV ingestion.
- **Normalization:** Tables are normalized to reduce redundancy and ensure data integrity.



3. CSV Data Loading

- **CSV Parsing:** Uses `CsvHelper` to read CSV files, mapping columns to entity fields.
- **Entity Management:**
  - Maintains dictionaries for unique Customers, Products, and Orders to avoid duplicates.
  - All IDs are treated as strings; no parsing to numeric types is performed.
- **Batch Insert:** Data is inserted into the database in batches for efficiency.

4. RESTful API Endpoints

- **UploadCsv:** Accepts CSV file uploads, parses data, and stores it into the database.
- **TopProducts:** Retrieves the top N products based on quantity sold within a date range.
  - Optional filters: **Category** and **Region**.
  - Returns product ID, name, category, and total quantity sold.

5. Analytical Queries

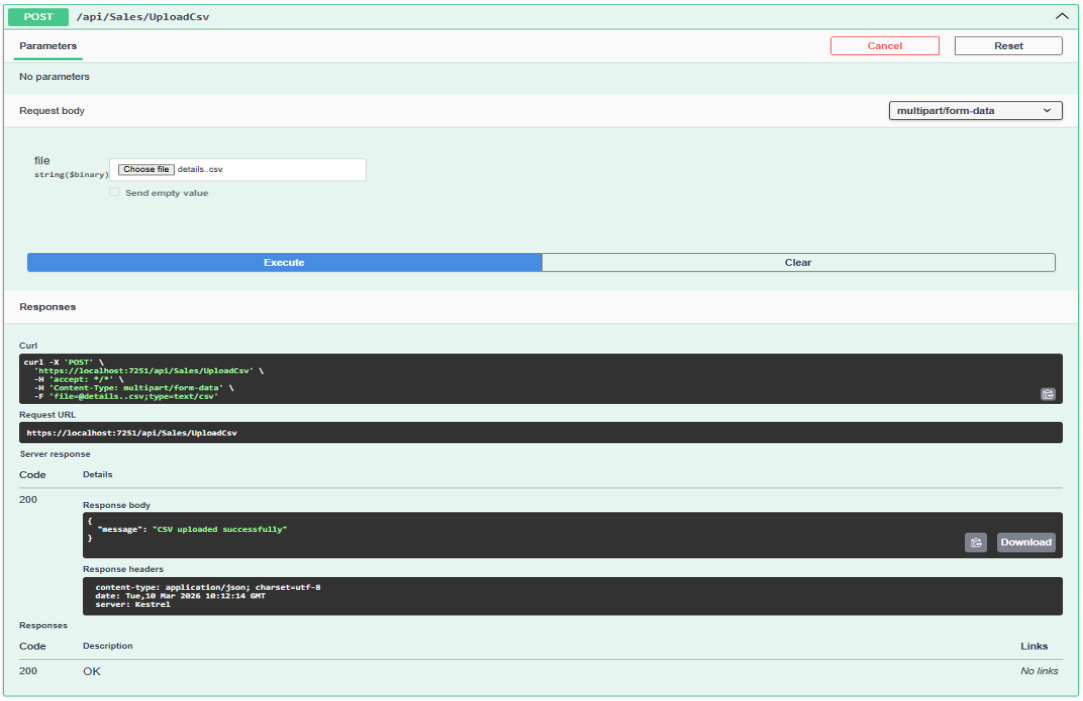
- Aggregates `OrderDetail` data grouped by Product.
- Supports filtering by category and region.
- Sorts by quantity sold to determine top-performing products.

6. Benefits of This Approach

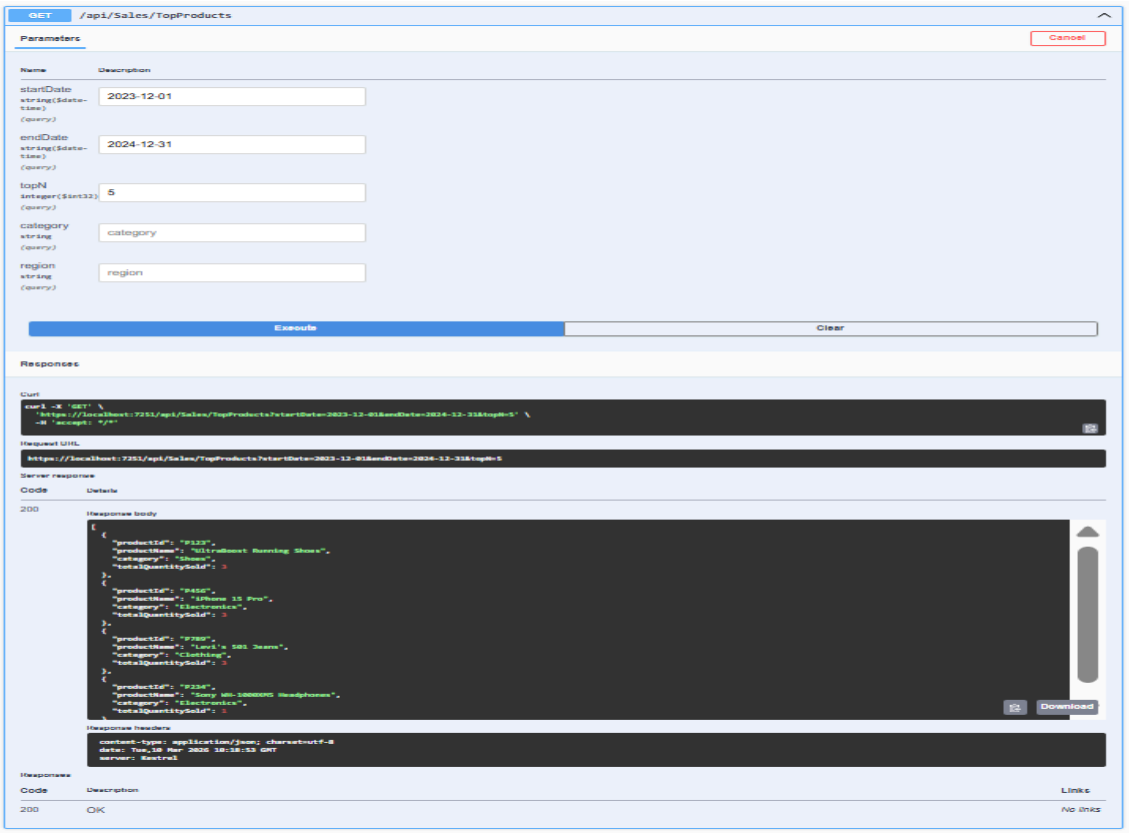
- **Scalable:** Can handle large CSV files efficiently.
- **Flexible:** ID fields are strings, eliminating parsing issues and simplifying data insertion.
- **Extensible:** Additional analytical endpoints can be added without schema changes.
- **Clean API Design:** Follows REST principles with clear separation of concerns.

Outputs :

1.Upload CSV **(Default Data)** :



2. Top N Products Overall **2023-12-01 - 2024-12-31** :



3.Top N Products By Category **(Electronics) 2023-12-01 - 2024-12-31 :**

GET /api/Sales/TopProducts

Parameters

Cancel

Name	Description
startDate string($date-time) (query)	2023-12-01
endDate string($date-time) (query)	2024-12-31
topN integer($int32) (query)	5
category string (query)	Electronics
region string (query)	region

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7251/api/Sales/TopProducts?startDate=2023-12-01&endDate=2024-12-31&topN=5&category=Electronics' \
  -H 'accept: */*'
```

Request URL

https://localhost:7251/api/Sales/TopProducts?startDate=2023-12-01&endDate=2024-12-31&topN=5&category=Electronics

Server response

Code	Details
200	Response body


```
[
  {
    "productId": "P456",
    "productName": "iPhone 15 Pro",
    "category": "Electronics",
    "totalQuantitySold": 3
  },
  {
    "productId": "P234",
    "productName": "Sony WH-1000XM5 Headphones",
    "category": "Electronics",
    "totalQuantitySold": 1
  }
]
```


Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue,10 Mar 2026 10:26:43 GMT
server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links

3.Top N Products By Region **(Asia) 2023-12-01 - 2024-12-31 :**

GET /api/Sales/TopProducts

Parameters

Cancel

Name	Description
startDate string($date-time) (query)	2023-12-01
endDate string($date-time) (query)	2024-12-31
topN integer($int32) (query)	5
category string (query)	category
region string (query)	Asia

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:7251/api/Sales/TopProducts?startDate=2023-12-01&endDate=2024-12-31&topN=5&region=Asia' \
  -H 'accept: */*'
```

Request URL

https://localhost:7251/api/Sales/TopProducts?startDate=2023-12-01&endDate=2024-12-31&topN=5&region=Asia

Server response

Code	Details
200	Response body


```
[
  {
    "productId": "P789",
    "productName": "Levi's 501 Jeans",
    "category": "Clothing",
    "totalQuantitySold": 3
  },
  {
    "productId": "P456",
    "productName": "iPhone 15 Pro",
    "category": "Electronics",
    "totalQuantitySold": 2
  }
]
```


Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue,10 Mar 2026 10:24:24 GMT
server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links